

Operator Discovery Report: restart_controller_stagnation_restart_perturb

- Operator type: restart_controller.
- Host scaffold: nearest_neighbor_2opt.
- Core idea: Deterministic restart growth tuned for transfer: triggers restarts on short stagnation, increases restart count gradually, and always applies the configured perturbation on restart to escape nearest-neighbor traps without
- Novelty classification: adaptive_restart_controller.
- Rediscovery signature: restart_controller:2:9:1.
- Surviving candidate: False.
- Problem class where it helps: clustered_tsp, two_cluster_bottleneck_tsp.

Pseudocode

- Start from a conservative restart budget.
- Increase restart count only after the configured stagnation trigger.
- Keep perturbation restarts bounded by a deterministic maximum.

Complexity

- Non-empty lines: 12.
- Complexity score: 0.44.
- Fingerprint: 6d23e789fdcd92f4.

Why It Should Help

- The operator changes restart pressure under stagnation, which is a compact way to regulate exploration without rewriting search internals.

Failure Cases

- Does not transplant cleanly into sparse_three_opt.
- Does not transplant cleanly into clustered_local_search.
- Runtime inflation is high at 0.770705.

Transplant Results

- nearest_neighbor_2opt gap delta 0.0, runtime delta 219.517629, distance-eval delta 5819.428571.
- cheapest_insertion_2opt gap delta -0.000426, runtime delta 890.979586, distance-eval delta 4118.285714.
- random_restart_2opt gap delta 0.0, runtime delta -197.947543, distance-eval delta -11440.0.
- sparse_three_opt gap delta 0.036727, runtime delta 6.196343, distance-eval delta 60.714286.
- clustered_local_search gap delta 0.014715, runtime delta -69.483086, distance-eval delta -6635.142857.

Instance-Family Results

- heldout_tsplib operator gap 0.235058 vs baseline 0.235058.

- clustered_tsp operator gap 0.083393 vs baseline 0.110853.
- elongated_corridor_tsp operator gap 0.003204 vs baseline 0.003204.
- grid_like_tsp operator gap 0.106696 vs baseline 0.106696.
- nearest_neighbor_trap_tsp operator gap 0.0 vs baseline 0.0.
- two_cluster_bottleneck_tsp operator gap 0.084632 vs baseline 0.109057.
- uniform_euclidean operator gap 0.0 vs baseline 0.0.

Pareto Results

- Gap delta: -0.002615.
- Runtime inflation: 0.770705.
- Distance-evaluation inflation: 0.84967.
- Same-gap-faster flag: False.
- Better-gap-same-runtime flag: False.

Ablation Results

- disabled mean gap 0.121317 and delta vs full operator 0.002615.
- fixed_restart_budget mean gap 0.118702 and delta vs full operator 0.0.
- wider_restart_budget mean gap 0.118702 and delta vs full operator 0.0.